

Requirements and Challenges

☐ Quality

- ☐ Ability to deal with different types of attributes: Numerical, categorical, text, multimedia, networks, and mixture of multiple types
- ☐ Discovery of clusters with arbitrary shape
- ☐ Ability to deal with noisy data

☐ Scalability

- ☐ Clustering all the data instead of only on samples
- ☐ High dimensionality
- ☐ Incremental or stream clustering and insensitivity to input order

☐ Constraint-based clustering

- ☐ User-given preferences or constraints; domain knowledge; user queries

☐ Interpretability and usability

Cluster Analysis: A Multi-Dimensional Categorization

❑ Technique-Centered

- ❑ Distance-based methods
- ❑ Density-based and grid-based methods
- ❑ Probabilistic and generative models
- ❑ Leveraging dimensionality reduction methods
- ❑ High-dimensional clustering
- ❑ Scalable techniques for cluster analysis

❑ Data Type-Centered

- ❑ Clustering numerical data, categorical data, text data, multimedia data, time-series data, sequences, stream data, networked data, uncertain data

❑ Additional Insight-Centered

- ❑ Visual insights, semi-supervised, ensemble-based, validation-based